

**Applied Thermodynamics
(ME2002)**

Date: November 2, 2024

Course Instructor(s)

Dr Huzaifa Rauf

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Sessional-2 Exam

Total Time: 1 Hours

Total Marks: 40

Total Questions: 02

Semester: F-2024

Campus: Lahore

Deptt: Electrical
Engineering

Student Name

Roll No

Section

Student

Vetted by

Vetter Signature

Instruction/Notes:

- Submit the question paper along with your answer sheet.
- Use valid assumptions if needed, don't ask any questions.
- Your answers should be correct up to three decimal places with appropriate units.
- Values read from tables should have the same number of significant figures as mentioned in the tables.
- You are allowed to bring two A4 sized cheat sheets, as well as thermodynamic tables

CLO #2: Analyze energy of a system using energy equation

Q1:

[marks 20] / 20

A steam turbine receives steam at its inlet at 3 Kg/s, 1000 Kpa, 400°C, at a height of 6m, and 40 m/s. The heat loss is 10KJ/Kg. Exit is at 100°C, at a height of 1m and 5 m/s as saturated vapor. Find out power produced and specific work done.

Now if the condenser at the output of the turbine improves its performance such that the exit state becomes 50°C, and 5 m/s as saturated vapor, keeping everything else constant. How much additional power is produced in the second case?

CLO #3: Analyze modes of heat transfer

Q2:

[marks =20]

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An electric water-heater has the shape of a perfect cylinder. The heating rods are present inside the heater. All the surface area of heater is covered up with insulation boards. The diameter of the water heater (excluding the insulation) is 0.2m and its length is 1.5m. The inner insulation surface is at 80°C and the outside surface is at 25°C and the board material has a conductivity of 0.09 W/m K. Find out thickness of insulation board required to limit the heat transfer loss to 200 W?

Assuming that the outside air temperature is 9°C find out heat lost through radiation. Also find out two different values of co efficient of convection ignoring heat radiation and not ignoring heat radiation.

